A Comparative Study of Deep Learning Models for Food Freshness Detection Using Transform Learning

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Abstract. The effectiveness of the deep learning models ResNet50, MobileNetV2, VGG16, InceptionV3, and EfficientNetB0 in identifying the freshness of food is evaluated in this study using visual analysis. In resource-constrained situations, these designs are ideal for automated food quality inspection and real-time freshness monitoring as they offer higher accuracy or computational efficiency. Transform learning are used to develop binary classifiers, which were then trained on a dataset of annotated food photos and assessed for efficiency and accuracy. After undergoing standardized preprocessing, the models' capacity to differentiate between fresh and stale food in a variety of test photos was evaluated. The results show the advantages and disadvantages of each model. The current stream of research frequently concentrates on generic picture classification tasks instead of the particular difficulties of distinguishing subtle visual differences in food freshness, leaving a gap in knowledge of model robustness under real-world conditions. The study advances deep learning applications in food quality evaluation by offering useful insights for model selection based on operational requirements.

Keywords: Deep Learning Models, Food Freshness Detection, Transform Learning

1 Introduction

In food manufacturing and supply chains, automated freshness detection using machine learning has grown in importance due to the increased focus on food safety and quality assurance. Conventional techniques for assessing freshness, including chemical analysis or sensory examination, frequently need specialized tools, skilled workers, and a significant amount of time. Because of these drawbacks, they are less appropriate for real-time, high-throughput applications in the contemporary food industry.

In order to classify food freshness directly from photos, this study uses five popular convolutional neural networks (CNNs) as feature extractors in a deep learning-based approach. The study uses ResNet50, MobileNetV2, VGG16, InceptionV3, and EfficientNetB0 to compare how well each model performs as a binary classifier for differentiating between stale and new food. Adapting these pre-trained models through transform learning approaches, they are then refined and assessed on a collection of photos with freshness annotations.

The main objective is to examine and contrast the performance of every CNN architecture with respect to classification accuracy, processing speed, and computational efficiency. The study looks at these variables in order to determine which model architecture provides the best balance between accuracy and resource usage, with the goal of supporting real-world applications where precise and quick freshness detection is crucial. The research findings offer valuable perspectives on the choice and refinement of CNNs for food quality control systems, which could lead to improvements in automated, dependable, and scalable freshness detection throughout the food supply chain.

2 Literature review

Logistics problems are a major factor in food loss in the fruit and vegetable supply chain (FVSC), according to an analysis of logistics-related food loss drivers (LRFLDs) by Surucu-Balci and Tuna [1]. To identify important LRFLDs, the review technique involved expert interviews and a rigorous literature analysis. Transportation and warehousing are frequently identified as the areas with the most effects on food loss because of issues including inadequate facilities and a lack of cooperation. To rank these drivers, analytical techniques like the Analytical Hierarchy Process (AHP) are frequently employed. The results show that poor packaging materials and a lack of coordination are significant sub-drivers. These findings serve as a foundation for tactics aimed at improving logistics and lowering food loss.

The research conducted by Karakaya et al. [2] examined the automatic classification of food freshness, which is important in the food sector, particularly for quick and

impartial rotting detection from the phases of manufacturing to consumption. Because traditional spoilage detection techniques are sometimes subjective and time-consuming, effective automation technologies are required. Histograms, gray level co-occurrence matrices (GLCM), bag of features (BoF), and convolutional neural networks (CNN) for feature extraction were employed in this study to categorize fresh and rotten samples of three different fruit varieties using image-based methodologies. According to experimental results, CNN-based features produced the best accuracy, with success rates of 96.72% for six-class classifications and 99.19% for three-class classifications. CNN's superior performance in multi-class setups was highlighted by the somewhat good performance of GLCM features and the relatively poor accuracy of BoF.

Li et al. [3] tackled food safety issues by highlighting food freshness as a crucial determinant and investigating intelligent packaging based on artificial intelligence (AI) as a cutting-edge freshness monitoring method. Even though intelligent packaging has advantages like cost-effectiveness and efficiency, its real-world applications are still somewhat limited. This analysis highlights the benefits and drawbacks of the intelligent packaging technologies already in use. Challenges include, in particular, sensor stability difficulties, processing demands of AI algorithms, and the massive data requirements of neural networks. Furthermore, the majority of food freshness detection uses neural networks and support vector machines (SVMs), which might produce inaccurate results in some situations and require a large number of training samples.

Liu and Grunert [4] investigated the way food-related beliefs affect total life satisfaction in their study of life satisfaction among the elderly in China. Their study, which was based on interviews with 452 people in Beijing who were over 60, reveals a significant positive association between overall life happiness and food-related life satisfaction—especially when compared to other nations. Three groups of senior consumers are identified: those who are moderately concerned (32%), those who are less concerned but more interested in enjoyment (29.4%), and those who are extremely concerned (38.6%). The two main factors influencing satisfaction are health and food safety, with cost considerations taking a backseat to safe and healthful eating. The necessity for safer and healthier food options to enhance the health of senior Chinese citizens is highlighted by this segmentation.

Fish freshness levels are categorized by Saputra et al. [5] utilizing image processing techniques, particularly color-based algorithms. Since traditional freshness inspections in markets rely on expert evaluations that can be physically taxing, they have become ineffective in light of rising fish consumption. This study uses RGB color data from fisheye photos to determine the freshness of fish using the Naive Bayes classification approach. RGB extraction, image cropping, segmentation, training, and testing are all part of the study process. There were 210 photos in the dataset, 147 for training and 63 for testing, and they were split into three categories: fresh, slightly rotting, and entirely rotten. The findings demonstrate that the Naive Bayes algorithm

has the potential to be a successful digital tool for market application, as it attained a 79.37% accuracy rate in classifying fish freshness.

Navotas et al. [6] developed an Android app that uses visual processing to categorize the freshness of common fish in the Philippines. Fish eyes and gills' RGB readings are analyzed by the app to determine shelf life and freshness levels. It was created using OpenCV and neural networks, and in regulated illumination, it produced precise results.

Pandey et al. [7] examined how machine learning (ML) can be used to optimize production, packaging, and safety procedures in the food business. By anticipating customer preferences, machine learning (ML) improves productivity, lowers waste, and makes tailored products possible. ML aids in the detection of contamination and enhances the use of ingredients in food safety. It optimizes storage conditions by assessing data and supporting nanotechnological improvements in food preservation. ML is also used in agriculture, where it helps with supply chain optimization and quality evaluation. By utilizing nanotechnology, machine learning (ML) enhances food distribution and packaging systems, guaranteeing quality and reducing the possibility of contamination, ultimately improving consumer pleasure and safety.

Valentino et al. [8] suggested a computer vision-based technique that use deep learning, more especially a Convolutional Neural Network (CNN), to assess fruit freshness in Indonesia, which has a tropical climate ideal for growing fruits and vegetables. Fruits that are unfit for consumption are frequently shipped by fruit vendors due to mistakes in sorting and packaging. Fresh and rotting fruit datasets from Kaggle were used to test the CNN model. This deep learning approach aims to enhance fruit quality by enhancing food deterioration detection from manufacturing to consumption.

Mukhiddinov et al. [9] used an enhanced YOLOv4 model to create a deep learning system for determining the freshness of fruits and vegetables. After determining the sort of object in an image, the algorithm determines whether it is fresh or rotting. By enhancing the YOLOv4 backbone with the Mish activation function, the model achieved higher precision, surpassing YOLOv4 and YOLOv3 with average precision scores of 50.4%, 49.3%, and 41.7%, respectively. Because of its high accuracy (97.6% for fruits and 97% for vegetables), the system has the potential to automate processes in the food business and help consumers choose fresh products, thereby reducing the risk of foodborne illnesses.

Yuan and Chen [10] researched pre-trained deep learning models, which are GoogleNet, DenseNet-201 and ResNeXt-101, with the Principal Dimensionally Analysis (PCA) reduction technique with the highest accuracy rate 96.98%. For research purposes, the dataset used 12000 images and images collected from different sources like Google Images, Bing Images, Kaggle, the Fruit360 dataset, and the Sriram R.K. dataset, which provided pure, fresh category samples.

3 Methodology

3.1 Model Selection

Five well-known architectures were chosen for the study based on their shown effectiveness in image classification tasks: ResNet50, MobileNetV2, VGG16, InceptionV3, and EfficientNetB0. These models are popular because of their many structural approaches, which provide a thorough assessment of their applicability, accuracy, and efficiency for real-time food freshness detection.

A. ResNet 50:
A deep CNN called ResNet50 (Residual Network with 50 layers) uses residual learning to tackle the problem of gradients diminishing. In an attempt to facilitate deeper network training without sacrificing efficiency, it uses shortcut connections, which are referred to as skip connections. ResNet50 efficiently captures the tiny characteristics and textures necessary to differentiate between fresh and stale food products thanks to its powerful ability to extract hierarchical features.

B. MobileNetV2:
A lightweight CNN designed for real-time and mobile applications is called MobileNetV2. It uses linear bottleneck layers and depth wise separable convolutions, significantly lower computing costs without impacting accuracy. MobileNetV2's effective architecture balances speed and precision, therefore being compatible with embedded food quality monitoring systems.

C. VGG 16:
The 16-layer VGG16 deep CNN has tiny 3x3 convolutional filters that improve feature extraction. VGG16's capacity to capture fine-grained spatial characteristics makes it useful for food texture analysis, despite its comparatively high computing cost. But compared to other contemporary alternatives, it uses more memory and computing resources.

D. Inception:
InceptionV3 is a powerful CNN that uses several filter sizes in a single layer and factorized convolutional to increase efficiency. Food photos with varied textures and structures may be effectively classified because of its inception modules, which enable it to assess many visual information scales at once. In the categories of accuracy and computing efficiency, this serves as a decent compromise.

E. EfficientNetB0:
EfficientNetB0 is a revolutionary CNN that uses compound scaling to maximize accuracy and efficiency. Depth, breadth, and resolution are all equally scaled to optimize performance at the lowest possible computational cost. Ef-

ficientNetB0 is perfect for large-scale food freshness classification jobs since it is lightweight and excellent at catching important visual patterns.

Using pre-trained weights from the ImageNet dataset, each model was set up to take advantage of transform learning. In order to modify the models for binary classification, two Dense layers were added to the top layers, followed by a Global Average Pooling layer. Food products were classified as either fresh or stale using a sigmoid activation function in the final Dense layer. With this configuration, each model may effectively use learnt picture attributes while concentrating on classifying the freshness of food.

3.2 Training and Evaluation

Binary cross-entropy loss, a popular metric for assessing classification tasks, was used to train the models for binary classification. This loss function has the following definition:

$$L = -\frac{1}{n}\sum_{x=0}^{M-1}\left(y_i \log(p_i) + (1 - y_i)\log(1 - p_i)\right) \quad \text{------ (1)}$$

where, p_i is the expected probability for the fresh class, and y_i is the actual class label (0 for stale, 1 for fresh). The models effectively learn to differentiate between fresh and stale food products by reducing this loss.

This method aids in gauging how well the models differentiate between two classes, in this case, fresh and stale food products. By encouraging the models to identify unique patterns linked to each class, binary cross-entropy training enables them to generate predictions that are more accurate when based on the dataset's visual attributes.

To classify the images, a prediction threshold of 0.5 was applied:

$$\text{Class} = \{\text{Fresh} \quad \text{if } p_i \geq 0.5, \text{ Stale} \quad \text{if } p_i < 0.5\} \quad \text{- - - - - - - - - - - - (2)}$$

To classify the images, a prediction threshold of 0.5 was applied. Values above 0.5 under this cutoff point denoted a "fresh" categorization, whilst values below 0.5 denoted a "stale." By offering a straightforward and reliable guideline for analysing the models' predictions, this threshold made it possible to distinguish between new and stale classes and made it easier to assess performance across various model architectures.

3.3 Methodology

Gathering a varied collection of food photos is the first step in the process, which implies the fact that fresh and stale foods are evenly represented. 50 images were used for testing purposes. Preprocessing techniques including scaling, resizing, and nor-

malizing are applied to these photos in order to preserve dataset integrity and enhance model performance. To improve model generalization and avoid overfitting, data augmentation techniques such as flipping, rotation, and brightness modifications are used.

Pre-trained Convolutional Neural Networks (CNNs), such ResNet50 and MobileNetV2, are configured with weights trained on picture datasets, like ImageNet, in order to take advantage of transform learning. In order to differentiate between fresh and stale food products, these models—which are renowned for their potent feature extraction capabilities—identify important visual patterns including texture, color, and structural integrity.

The model can capture important features that distinguish food quality according to the feature extraction procedure, which eliminates the need for onerous human feature engineering. Next the contrary, the extracted characteristics are improved and categorized by going through completely linked layers. In order to enhance classification accuracy and adjust to actual differences in food look, the models are refined using domain-specific food photos.

Loss functions, like categorical cross-entropy, direct the optimization process during training in order to reduce prediction errors. Underfitting and overfitting are avoided by adaptive learning rate techniques, which guarantee optimal convergence. Accuracy, precision, recall, and F1-score are among the evaluation criteria used to gauge model performance and guarantee resilience across various food categories.

After training is finished, test data that mirrors real-world situations is used to validate the model. After that, a trained model is used for prediction, processing fresh food photos and classifying each item as either "Fresh" or "Stale." The last classification layer generates predictions by giving each category a probability score.

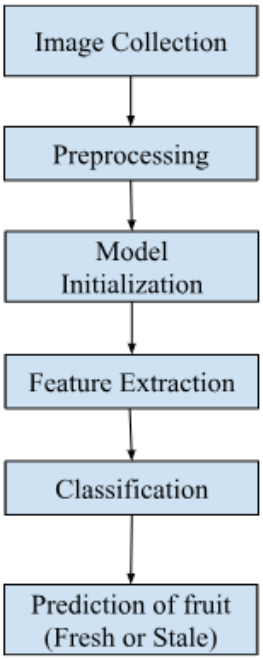


Fig. 1. General Steps of Experimentation

4 Results

The study's rotten test picture samples, exhibiting suggest varying levels of degradation across food categories, are shown in Figure 2. These images emphasise significant visual indicators of diminishing freshness, with the value colour shifts, texture deterioration, and the growth of mould. The algorithms' capacity for discerning between fresh and stale food was thoroughly assessed by examining these samples.



Fig. 2. Rotten Fruit images

ResNet50 exhibits great performance on stale data (62.93%) and performs best on new data (89.46%). VGG16 performs better on stale data (64.98%) than fresh data (55.89%), but MobileNetV2 struggles with stale data, falling to 30.40%. Both fresh

and stale data demonstrate modest performance from InceptionV3 and Efficient-NetB0, with EfficientNetB0 experiencing a minor decline in both scenarios.

Models	**Fresh (%)**	**Stale (%)**
ResNet50	89.46	62.93
MobileNetV2	66.62	30.40
VGG16	55.89	64.98
InceptionV3	67.99	50.50
EfficientNetB0	55.64	48.78

Table 1. Accuracy (%) across several models

Prediction Accuracy: The prediction results from each model are displayed, showing the proportion of correct predictions. Studies show that model performance varies, with some designs demonstrating exceptional precision and others achieving a compromise between computational efficiency and accuracy. These results offer important information concerning how to choose the best model depending on the requirements of a certain application.

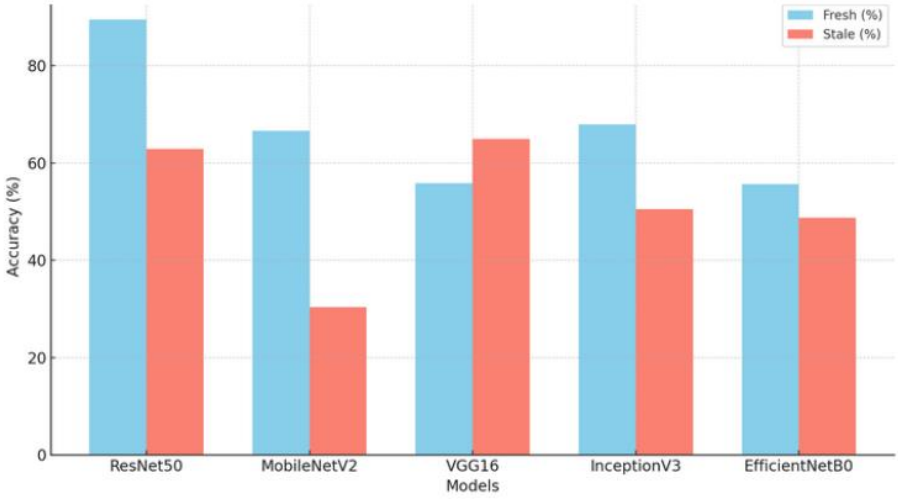


Fig. 3. Fresh and Stale Prediction Accuracy by Model

5 Conclusion

Deep learning models show immense potential for assessing the freshness of food, ResNet50 delivers the best accuracy, but on stale data, it remains stable. In comparison to MobileNetV2, VGG16 performs better on stale data. Reliability across a variety of circumstances is demonstrated by the steady but modest performance provided by InceptionV3 and EfficientNetB0. The two most successful models for automated food quality evaluation are ResNet50 and EfficientNetB0. In the future, real-time implementation, data from specific domains improvement, and extending categorization to various freshness levels should be the most important fields of exploration. The efficiency and accuracy of commercial food monitoring systems could potentially be boosted through these improvements. Automated food safety inspections might be made even more efficient by integrating these models with Internet of Things-based solutions. The robustness and generality of the model could be enhanced with a larger dataset that includes a wider range of food kinds and environmental circumstances.

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